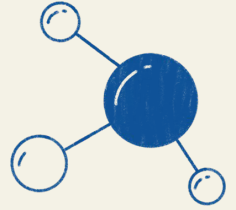
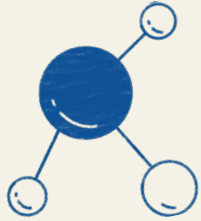


AlphaFold-guided redesign of a plant pectin methylesterase inhibitor for broad-spectrum disease resistance

Rohin Shanker

BIOE 190 Structural Bioinformatics





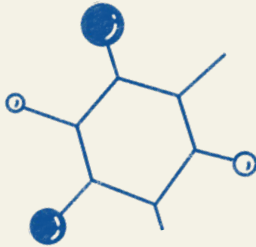
Introduction/Background

4th leading crop

produced globally by volume, according to the USDA

10-20% loss

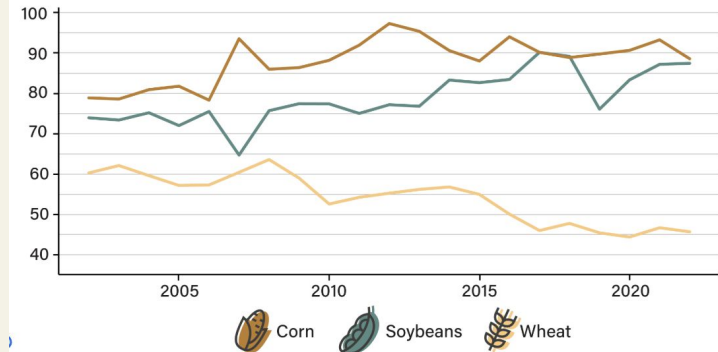
Amount of soybean crop lost due to disease



U.S. planted acreage of soybeans, corn, and wheat, 2002-2022

USDA Economic Research Service
U.S. DEPARTMENT OF AGRICULTURE

Acres (millions)



Source: USDA, Economic Research Service using data from USDA, National Agricultural Statistics Service, Crop Production publication.

Problem Context - Soybean root rot is a cell-wall problem

- Phytophthora sojae (oomycete) causes soybean root and stem rot, and can even overcome resistant cultivars
- The plant cell wall is an active layer of innate immunity
- Can wall pectin chemistry can be engineered for resistance?



Biological mechanism (cell wall + pectin system)

HG methylesterification defines the state of the pectin barrier

- HG is the dominant pectic polymer in the primary wall and middle lamella
- PME remove methyls and change wall chemistry
- PMEI proteins can preserve a high-methylesterification state
- dbCAN-based scan identified many pectin degrading enzymes in *P.sojae*

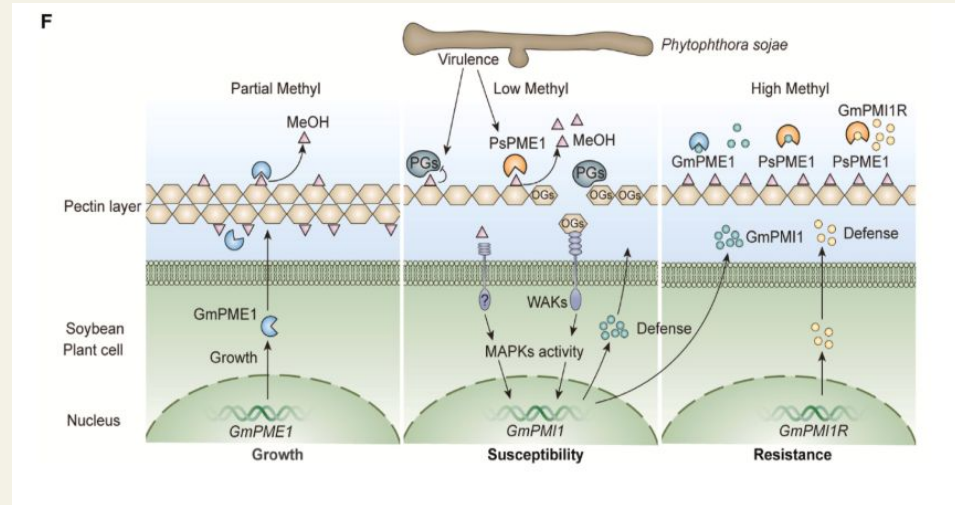
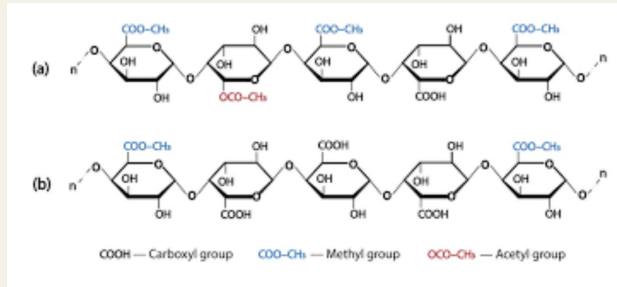
Important Vocab

- Homogalacturonan (HG), makes up ~65% of pectin - Oligogalacturonide (OG), short pectin fragment
- Pectin Methylesterase (PME) , removes pectin methyls to prepare pectin for degradation
- Pectin Methylesterase Inhibitor (PMEI), inhibits activity of PMEs
- GmPME1 = host PME used in normal plant activity
- GmPMI1 = host PMEI used to inhibit both host and infectious PMEs
- PsPG1 = Polygalacturonase, cleaves α -1,4 linkages between galacturonic acid residues in HG
- PsPME1 = Invader PME

Biological mechanism (cell wall + pectin system)

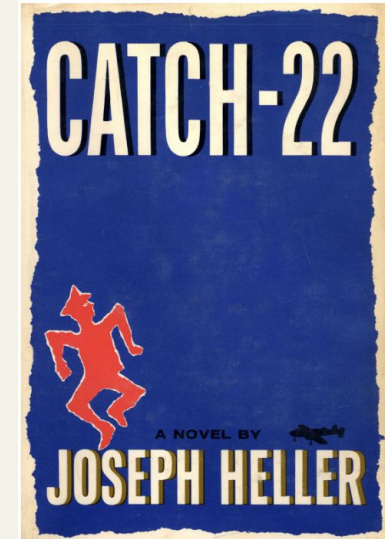
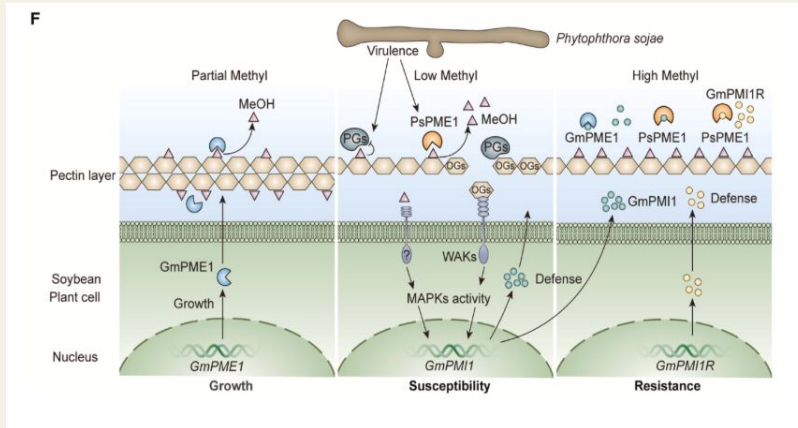
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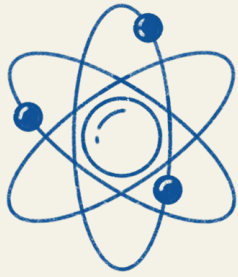
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Growth-Defense Tradeoff → Control Point

- High HG methylesterification protects the wall but prevents growth (Nguyen et. al, 2017)
- Low HG methylesterification allows growth but enables PsPG1-driven degradation
- Specificity problem





Study Goals

1. **Biology question:** does HG methylesterification control susceptibility?
2. **Mechanistic question:** what proteins (pathogen AND host) control it?
3. **Design question:** can a host inhibitor be made pathogen-selective?

Goals, Methods, Results



01

Establish wall remodeling during infection

05

GmPMI1 is induced as a host countermeasure

02

Identify necessity of PsPME1

06

AlphaFold-guided redesign of GmPMI1R

03

Identify PsPME1-PsPG1 synergy

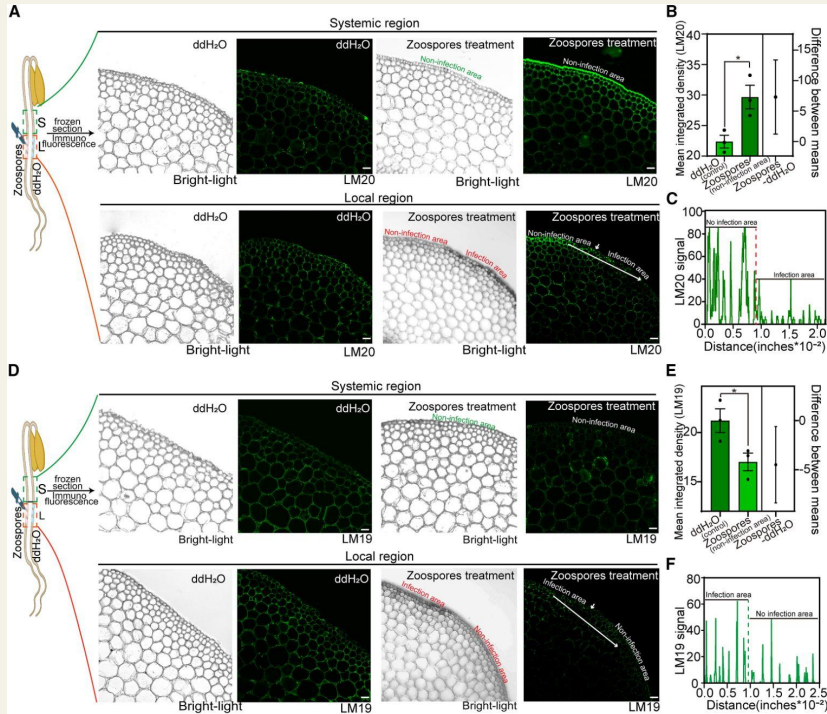
07

Resistance outcomes of GmPMI1R

04

Methanol activates defense signaling

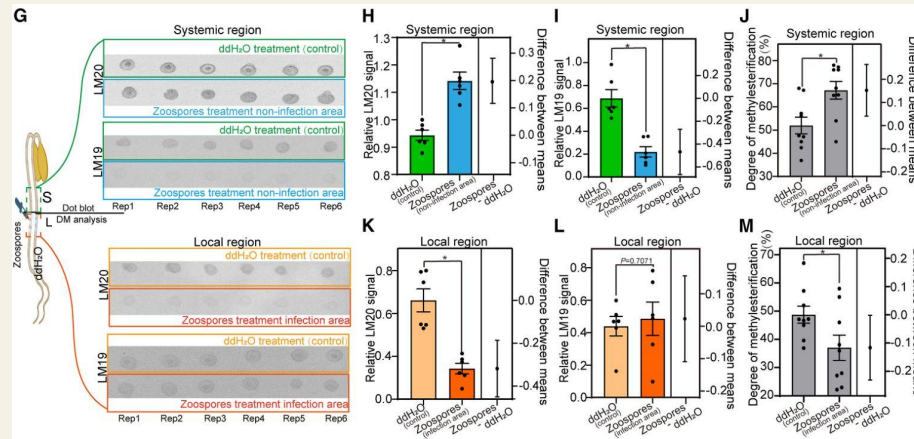
Figure 1 (1/2) Establish wall remodeling during infection



Goal: Show that pectin methylesterification is remodeled during early infection

- Immunofluorescence Imaging of infected hypocotyls (6 hpi)
 - LM20: High methylesterification
 - LM19: Low methylesterification
- Region types
 - Local = direct pathogen contact
 - Systemic = distant host response
- Distance traces along white line

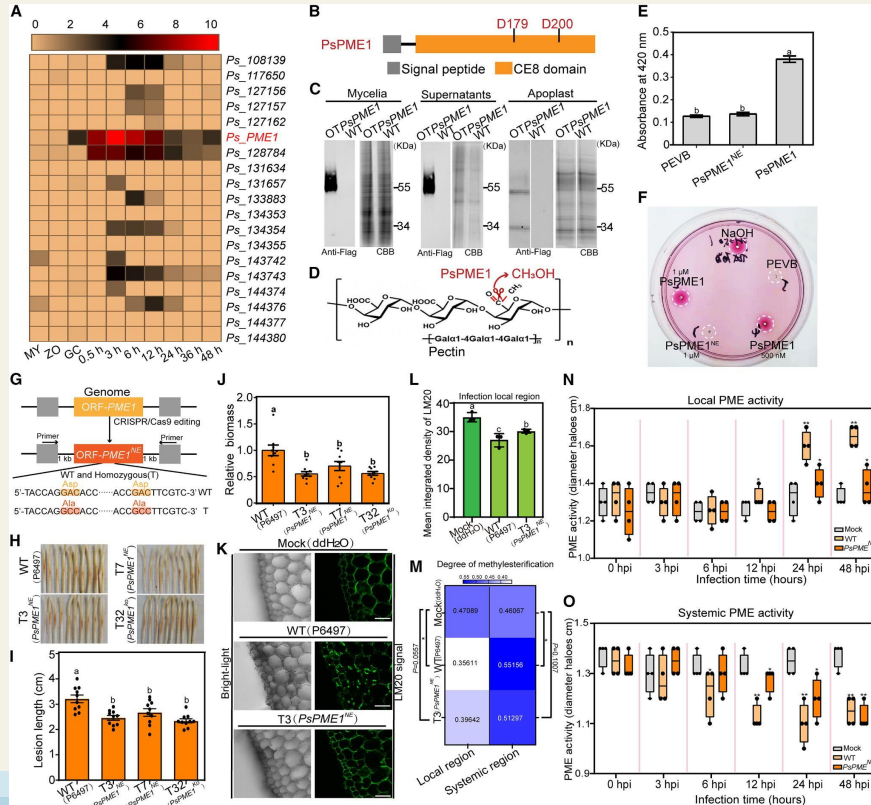
Figure 1 (2/2) Establish wall remodeling during infection



- Dot blot (6 hpi)
- LM20 → high ME, LM19 → low ME
- ~15% systemic increase and ~10% local decrease in degree of methylesterification (DM)

Key Results: Spatial heterogeneity → active remodeling by host + pathogen

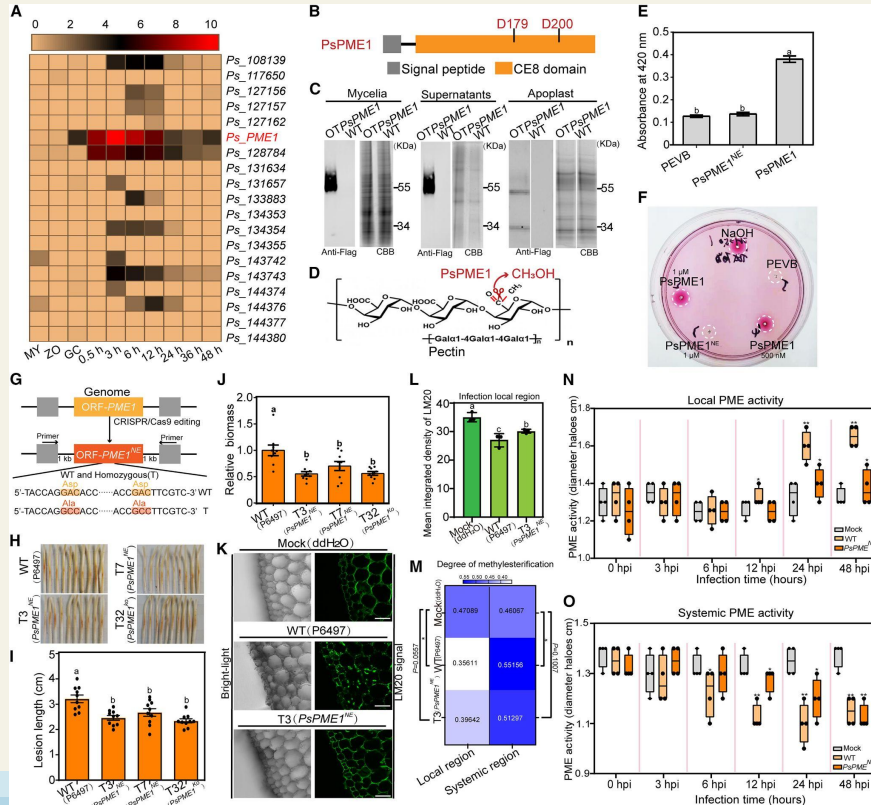
Figure 2 (1/2) Identify necessity of PsPME1



Goal: ID enzyme responsible for pectin demethylesterification

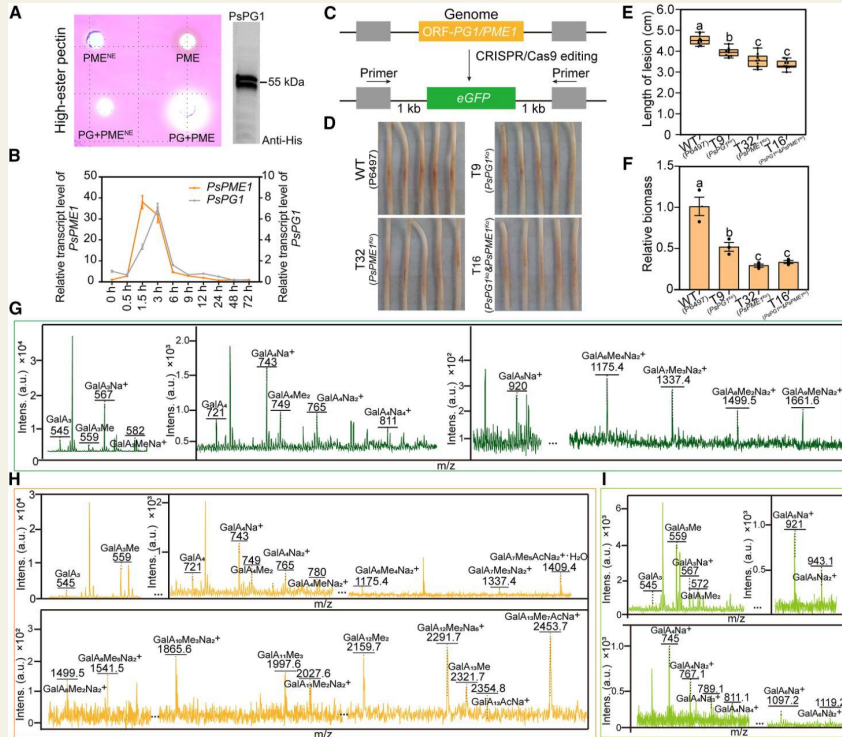
- Candidate ID + secretion / protein characterization
 - Found upregulated protein w/ CE8 domain
 - Immunoblot + FLAG tagging found it secreted
- MeOH Release + PECTOPLATE
 - Demonstrate that enzyme actually cleaves methylesters

Figure 2 (2/2) Identify necessity of PsPME1



- Functional genetics + virulence testing (WT, NE, KO)
 - lesion formation
 - pathogen biomass
 - wall-state effects in infected hypocotyls
- Key Results:** PsPME1 catalytic activity is important for full virulence

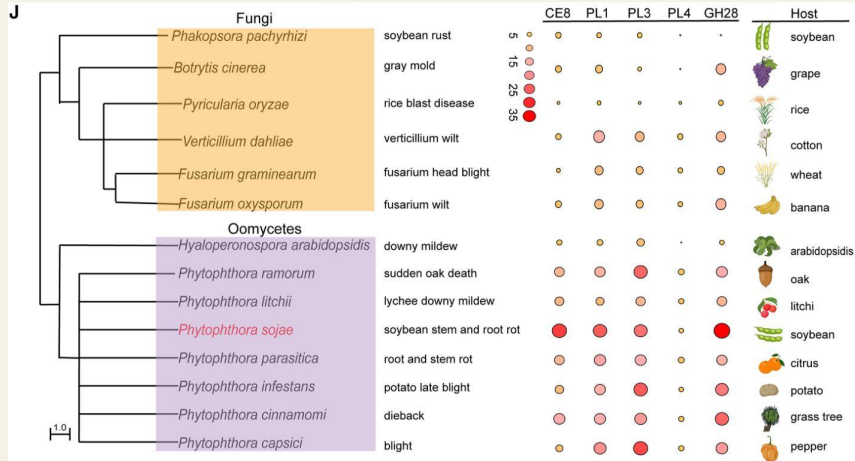
Figure 3 (1/2) Identify PsPME1-PsPG1 synergy



Goal: Test synergy between PsPME1 and PsPG1 in driving full infection

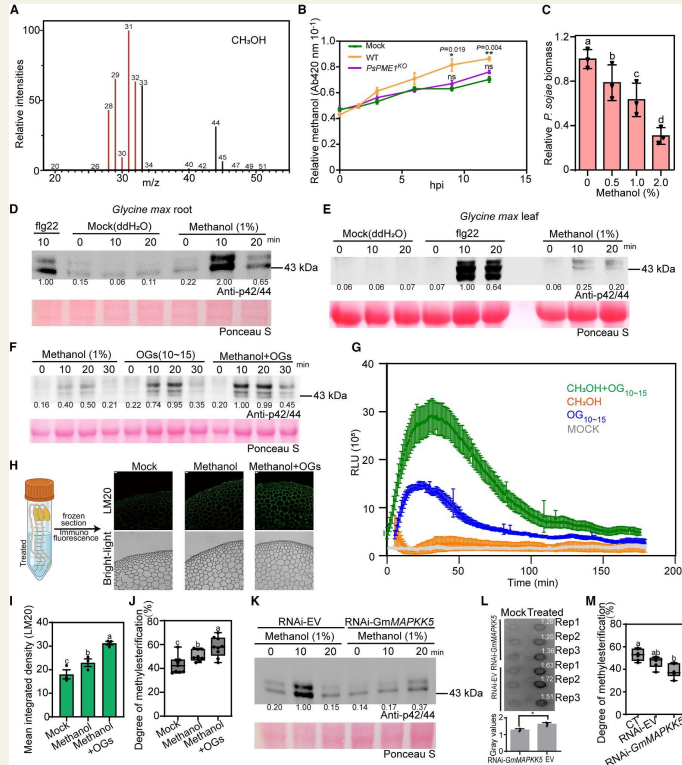
- PECTOPLATE + qRT-PCR suggest cooperation
- Validate with 1x KO and 2x KO variants and their expression phenotypes
- MALDI-TOF mass spectra on ddH₂O, WT, and 2x KO reveal Oligogalacturonide profile

Figure 3 (2/2) Identify host-side countermeasure



- Many pathogens encode pectin-degrading enzymes
- Result:** PsPME1 demethylesterifies HG, which allows PsPG1 to cleave homogalacturonan

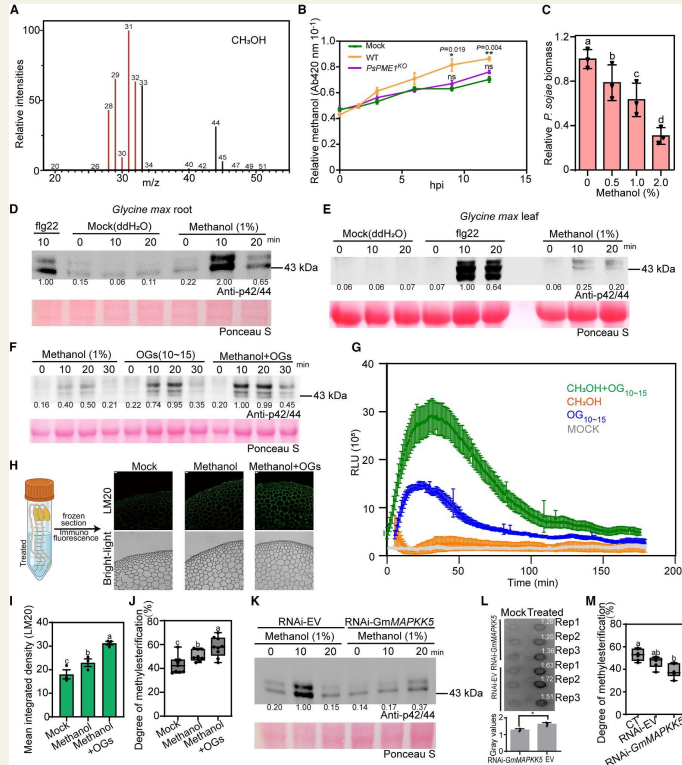
Figure 4 (1/2) Methanol activates defense signaling



Goal: Demonstrate that MeOH acts as a DAMP (damage associated molecular pattern) to promote a protective wall state

- GC/MS + Colorimetric assay to determine if PsPME1 produces MeOH
- Immunoblot to determine MeOH + OG₁₀₋₁₅ → MAPK → immune response (ROS) + methylesterification

Figure 4 (2/2) Methanol activates defense signaling

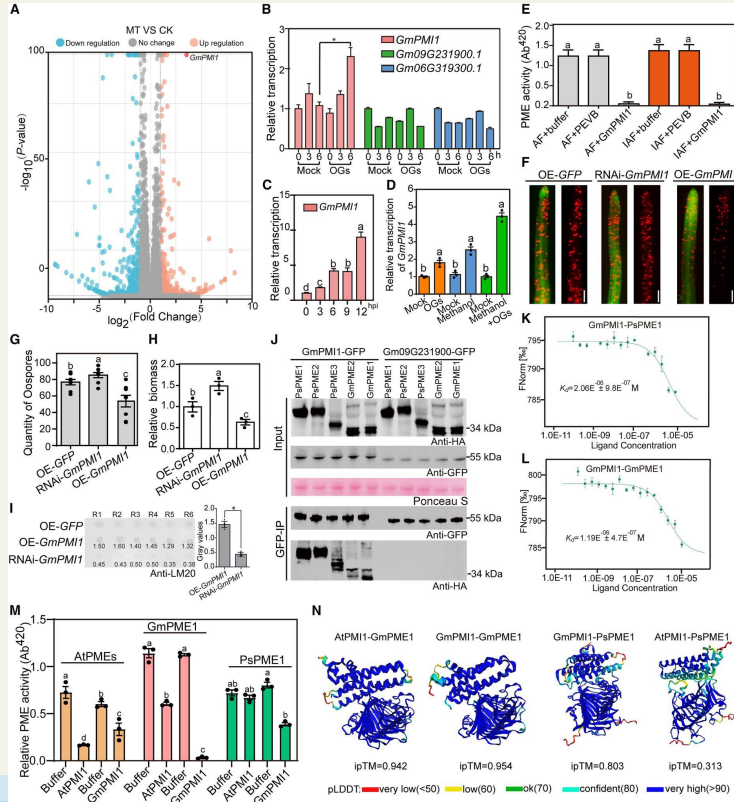


Result:

- Infection produces MeOH (partly via PsPME1)
- MeOH activates MAPKs and enhances OG₁₀₋₁₅-triggered MAPK + ROS responses
- MeOH increases methylesterification, partly through MAPK signalling

Figure 5 (1/2)

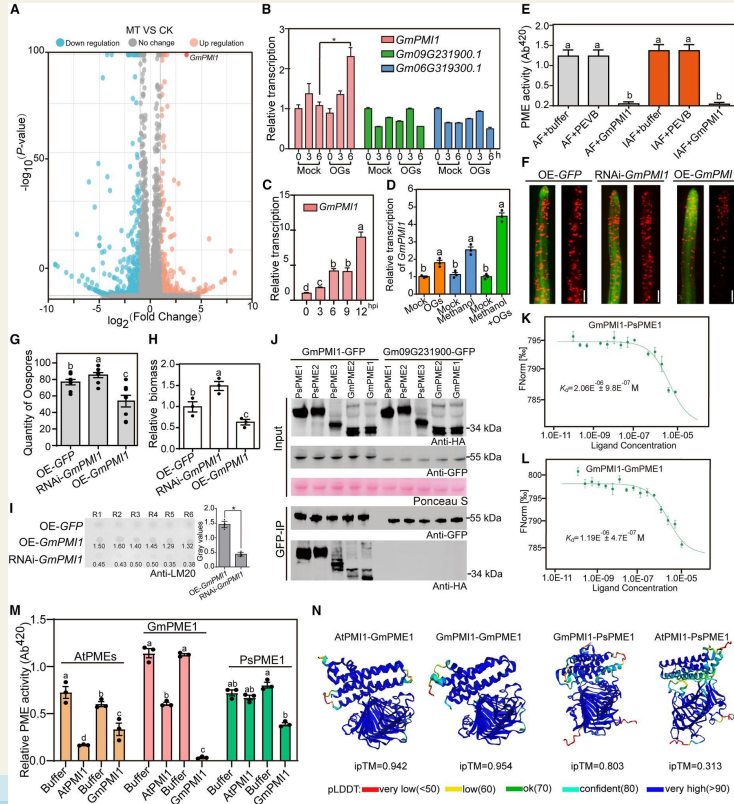
GmPMI1 is induced as a host countermeasure



Goal: Demonstrate that GmPMI1 is induced by infection and test defense capabilities

- RNA-seq + qRT-PCR on treated roots shows GmPMI1 ramp up
- PME activity assay + fluorescence microscopy show that GmPMI1 inhibits PME activity and stifles infection

Figure 5 (2/2) GmPMI1 is induced as a host countermeasure

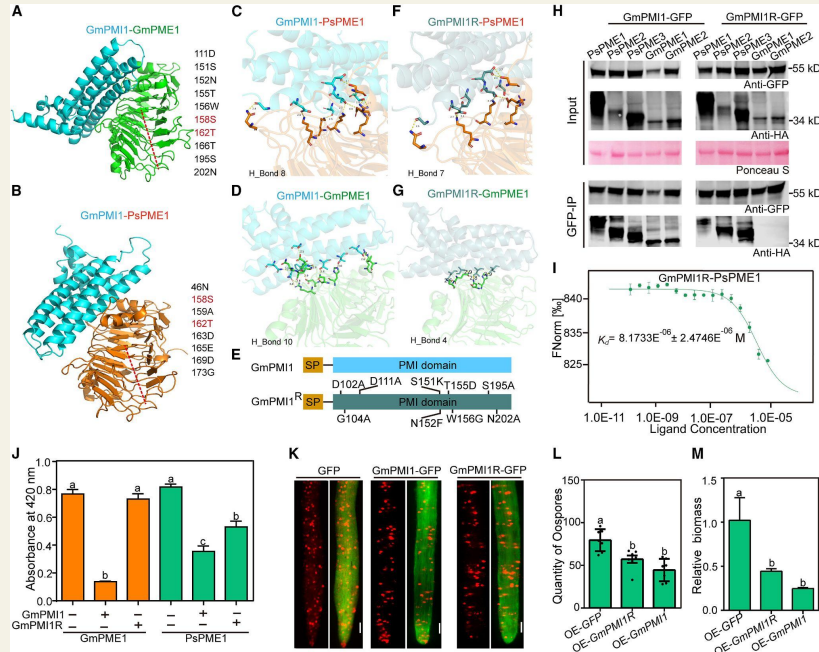


- co-IP, MST, colorimetric enzyme assay, AlphaFold2-Multimer simulation demonstrate non-specificity

Result:

- GmPMI1 induced by infection
- GmPMI1 has dual-inhibitory function

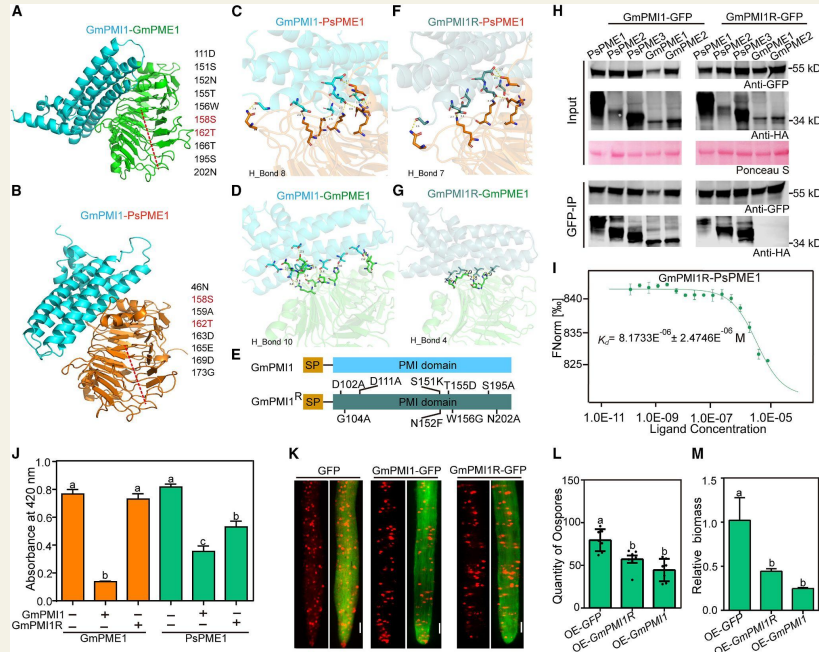
Figure 6 (1/2) AlphaFold-guided redesign of GmPMI1R



Goal: Show that GmPMI1 can be redesigned to be more pathogen selective

- Compare predicted interfaces and H-bonds
 - 10 GmPMI1-GmPME1
 - 8 GmPMI1-PsPME1
- Selected 9/10 for mutation
- Replaced all with alanine, then introduced substitution from NtCIF for stability

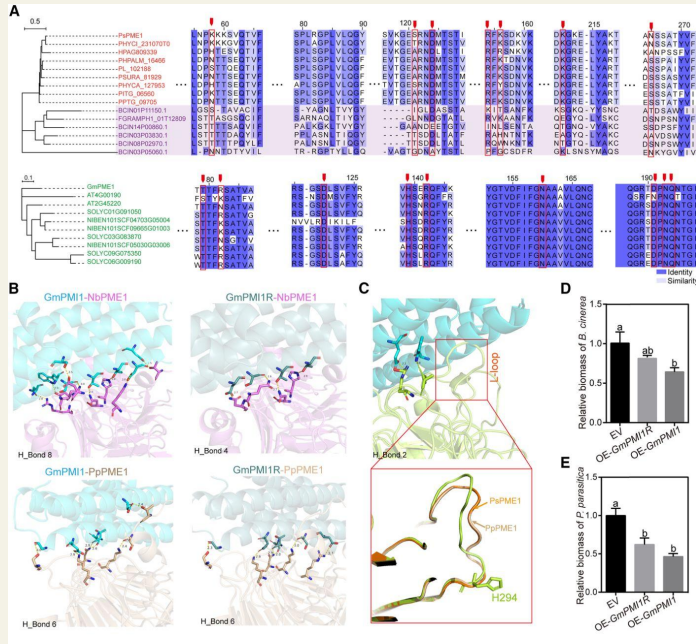
Figure 6 (2/2) AlphaFold-guided redesign of GmPMI1R



- Verified with AlphaFold2-multimer, co-IP, MST, and colorimetric enzyme assay
 - Tested in vivo with modified soybeans
- Result:** GmPMI1R showed high specificity to non-plant PMEs but was weaker than GmPMI1

Figure 7 (1/2)

Resistance outcomes of GmPMI1R



Goal: Test whether GmPMI1(R) specificity generalized across species

- Evolutionary sequence comparison to predict compatibility
- Modeled compatibility with PME from *N. Benthamiana* and *P. parasitica*
- Modeled and tested resistance to *B. cinerea* and *P. parasitica*
- Does not work for fungi

Discussion



Pathogen Strategy

P. sojae uses PsPME1 +
PsPG1 to weaken the *G. max*
cell walls



Growth-Defense Tradeoff

Broad inhibition in the plant
creates a growth cost



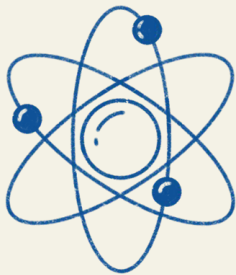
Inhibition as Defense

G. max uses GmPMI1 to
preserve wall
methylesterification



AlphaFold-Guided Redesign

Creating higher-specificity
PMIs can supercede this
tradeoff



Proposal

Gap

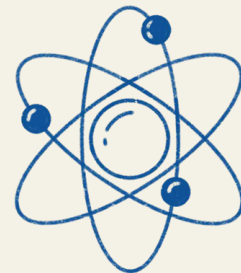
The paper designed a PMEI to specifically target one pathogen PME and specifically not interact with one host PME

- Can a PMEI be designed systematically and optimized to specifically target as many pathogen PMEs and as few plant PMEs?

Proposed Direction

- Model PME–PMEI across many different pathogens and plants
- Determine a subset of PMEIs that would be good starting points based on certain metrics (e.g. host-pathogen specificity)
- Iteratively redesign and test new proteins to optimize PMEIs
- Verify in vivo + vitro (infection outcomes)

Specific Aims



01

Identify PMEIs with strongest selective binding profiles

- Model PME–PMEI complexes using AlphaFold2-Multimer
- Compare interfaces across host and pathogen PMEs
- Identify most promising candidates for modification

02

Optimize structure-guided redesign of top PMEI scaffolds

- Using sequence interaction data, perform amino acid substitutions to improve specificity
- AlphaFold-Multimer, AlphaFold, co-immunoprecipitation,

Impact

- Ideally, stronger pathogen-specific inhibition without affecting host function
 - Lays groundwork for engineering more resistant crops
- Proof of concept for modifying enzyme function
- Establish pathogen PME resistance profiles for various plants

References

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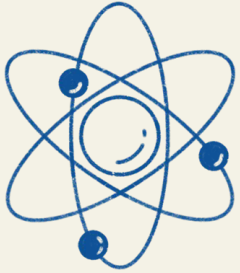
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Nguyen, H.P., Jeong, H.Y., Jeon, S.H., Kim, D., and Lee, C. (2017). *Rice pectin methylesterase inhibitor28 (OsPMEI28) encodes a functional PMEI and its overexpression results in a dwarf phenotype through increased pectin methylesterification levels.* Journal of Plant Physiology 208:17–25.



Questions?

